

Adam Petrovic

Senior Software Engineer – Distributed Systems, Cloud Infrastructure & Reliability

Sydney, Australia

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Summary

Senior software engineer with 14+ years building cloud platforms, object storage, and reliability systems, including 10+ years at Atlassian. Current work is on Atlassian's internal S3-compatible storage service: GCP migration, Kubernetes, storage analytics, incident response, and cost/reliability work. Earlier roles covered observability systems, a \$2M+ metrics migration, 180M metrics/minute pipelines, synthetic monitoring, and Jira Cloud SRE. Enjoys turning complexity and risk into RFCs, production code, and tooling adopted by other teams across the organisation.

Skills

- **Languages:** Kotlin, Go, Python, Java, TypeScript, JavaScript, SQL, Bash
- **Cloud & Infrastructure:** AWS, GCP, Kubernetes, Docker, Spinnaker, Flux, Helm, Kustomize, Talos Linux, CI/CD
- **Storage & Data Systems:** S3, GCS, DynamoDB, PostgreSQL, MySQL, Redis, sharding, backup/restore
- **Reliability & Observability:** SignalFx, Datadog, Prometheus, Grafana, Splunk, StatsD, Sentry, synthetic monitoring, SLOs
- **Data & Performance:** Kafka, Apache Flink, SQS, Databricks, Athena, Locust, high-volume telemetry pipelines

Experience

Atlassian (10+ Years)

Senior Engineer, Transactional Data Platform

Sydney, Australia

August 2023 – Present

- Led GCP migration for Atlassian's internal S3-compatible storage service across compute, credentials, data-integrity checks, and deployment readiness, delivering a critical milestone four weeks early.
- Designed Kotlin architecture for S3-compatible storage APIs, data residency, encryption, backup/restore, GCS integration, and multi-shard deployment paths for cloud-agnostic operation across AWS and GCP.
- Wrote RFCs for GCS checksum handling, reducing control-plane dependencies, and seekable compression, turning reliability, cost, and migration trade-offs into clear storage direction.
- Built S3 Inventory and DynamoDB export ingestion into the data platform, giving service owners storage cost attribution, object verification, and self-service analytics over multi-terabyte datasets.
- Replaced slow storage investigations with self-service queries, cutting 24-hour verification jobs to 5-minute checks and safely enabling \$25K/month object-tag savings across production buckets.
- During a high impact incident, built analysis pipelines under pressure to join database exports with S3 inventory, prove no data loss, and start continuous object verification for the platform's top risk.
- Created deployment-descriptor diff pipeline in Go/Bitbucket that flags risky infrastructure changes before merge, adopted by multiple platform teams to catch blast-radius issues before production.
- Led sharding work that split the storage service into isolated deployment units, delivering an isolated-cloud shard three weeks early and a repeatable rollout method for future shards and regions.
- Built repeatable Locust load tests for staging, letting engineers validate feature changes under production-like traffic, compare results across runs, and build confidence before deployment.

Engineering Manager, Observability

October 2020 – August 2023

- Led Better Metrics migration from Datadog to SignalFx for Atlassian engineering, saving more than \$2M annually while keeping critical metrics workflows stable for thousands of engineers during transition.
- Managed observability engineers through vendor evaluation, roadmap planning, stakeholder communication, incident reviews, hiring, coaching, performance development, and operational trade-offs.

- Led delivery of an HTTP routing layer between metrics aggregators and upstream observability sinks, using Atlassian's open-source GoStatsD project to make the Datadog-to-SignalFx migration safer at scale.

Senior Engineer, Observability

June 2017 – October 2020

- Built and operated Atlassian's metrics infrastructure at 180M datapoints/minute, supporting production monitoring, alerting, and incident diagnosis across cloud services and internal platforms.
- Developed Pollinator, a Go-based synthetic monitoring and post-deployment verification platform used across regions to catch reliability and performance regressions in release flows before customers did.
- Extended GoStatsD with Prometheus-style histograms and moved telemetry toward Kafka/Flink, improving percentile metrics, stream-processing headroom, and SLO diagnosis for service teams.

Senior Site Reliability Engineer, Jira Cloud

March 2015 – June 2017

- Led a cross-continental SRE group delivering Jira Cloud's European infrastructure ahead of a public launch milestone, supporting enterprise customers, regional growth, and product commitments.
- Reworked Jira Cloud site imports and introduced Wheel of Misfortune training, reducing failed customer imports, on-call toil, and incident-response gaps through rehearsed failure scenarios.

Freelancer.com (3.5 Years)

Sydney, Australia

Software Engineer

August 2011 – February 2015

- Built platform services for authentication, messaging, notifications, email, analytics, and fraud systems in a global marketplace using Python, Go, Redis, RabbitMQ, Thrift, and async workers.
- Helped move critical paths away from PHP monolith patterns into service-oriented APIs and async processing, improving request latency, failure isolation, and delivery speed for core marketplace flows.

Education

The University of Sydney

Sydney, Australia

Bachelor of Information Technology, Computer Science

2008 – 2012

- Dean's List with a distinction average.